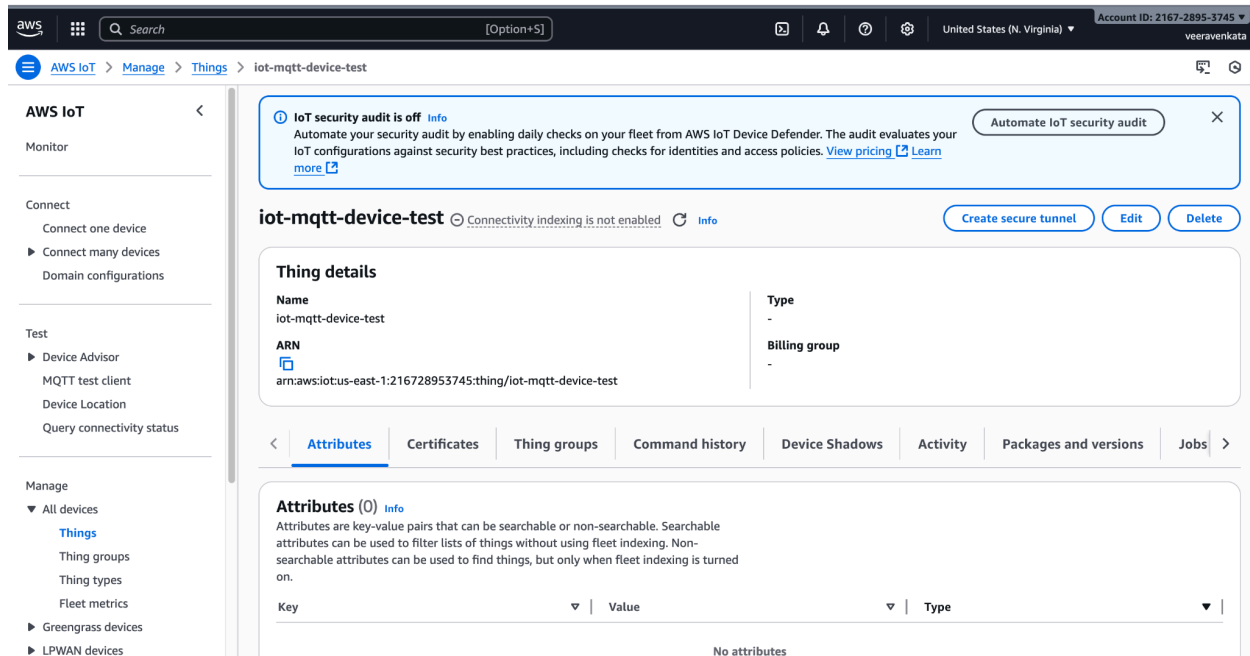


In-Class: AWS IoT Core

STEP 1 – Choose Device Option

We selected the virtual device on a local computer using the AWS IoT Device SDK for Python, which allows simulation without physical hardware or EC2.

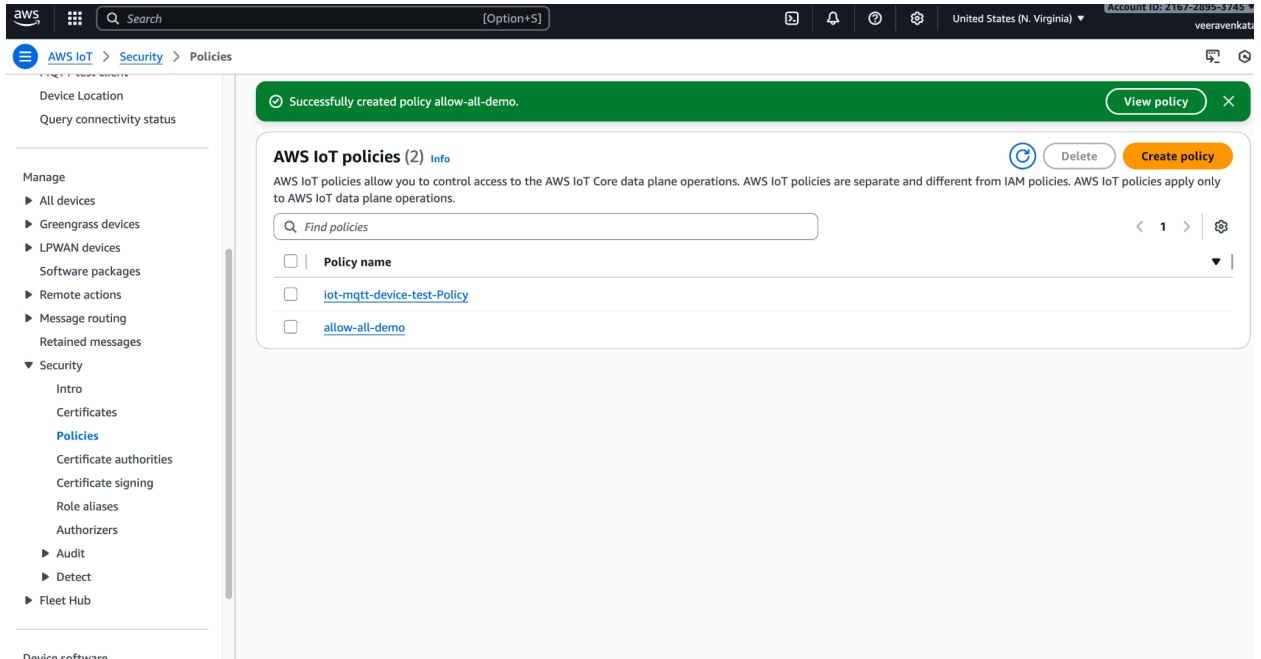


STEP 2 – Create AWS IoT Resources

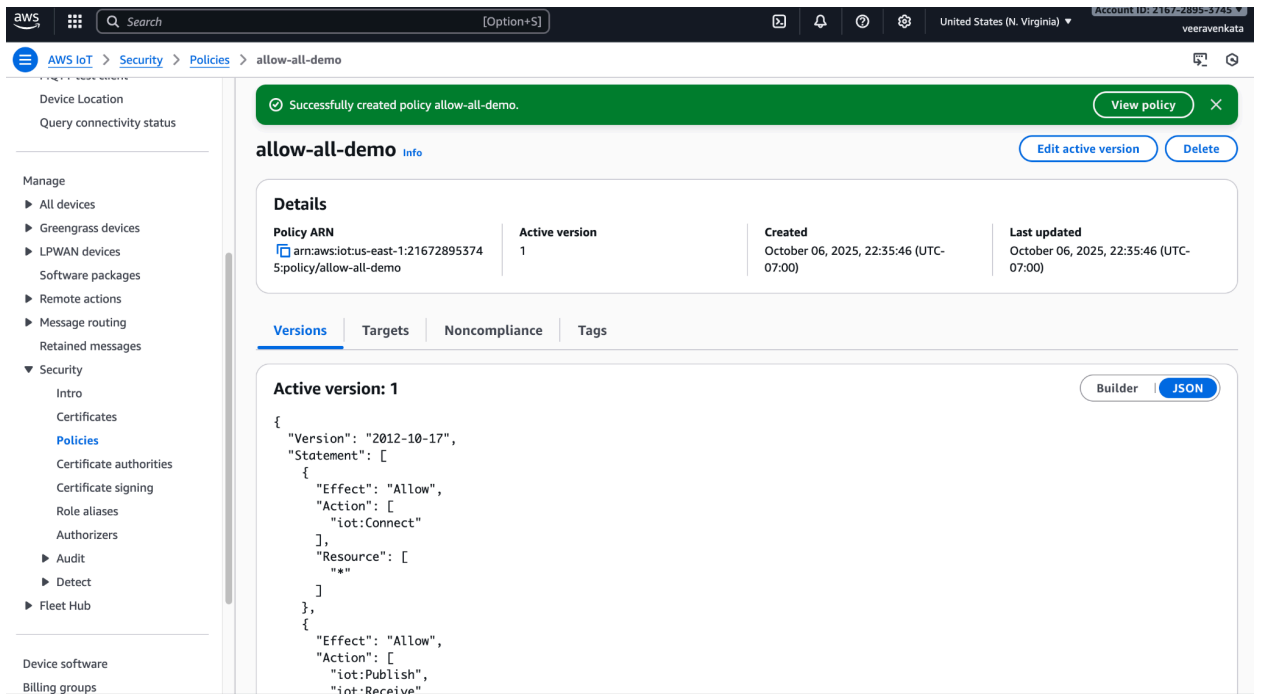
AWS IoT Policies

We created a Thing named `iot-mqtt-device-test`, downloaded the certificates, and created a policy named `allow-all-demo`.

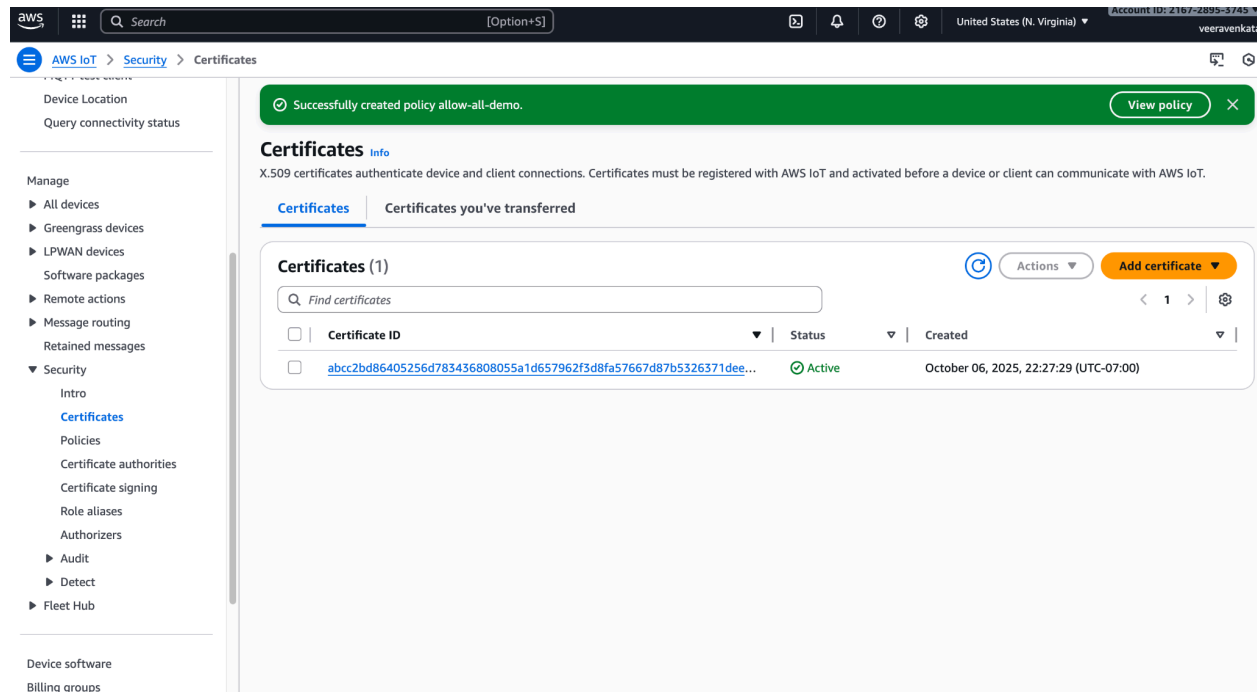
We attached both the Thing and the policy to the certificate and noted the device endpoint for later use.



Security Policy Json Configuration:



Certificates:

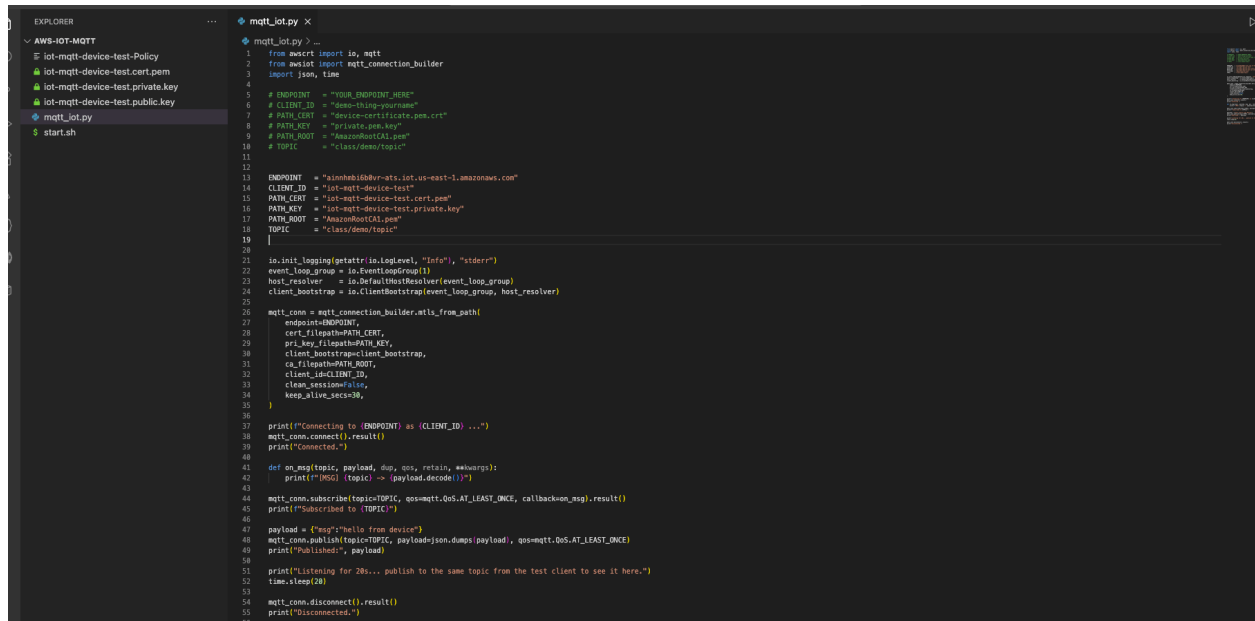


STEP 3 – Configure the Virtual Device (Python SDK)

We set up a virtual Python environment, installed the **AWS IoT Device SDK**, and configured our local script (`mqtt_iot.py`) with the correct endpoint and certificate files. The device successfully connected to AWS IoT Core and published a test MQTT message.

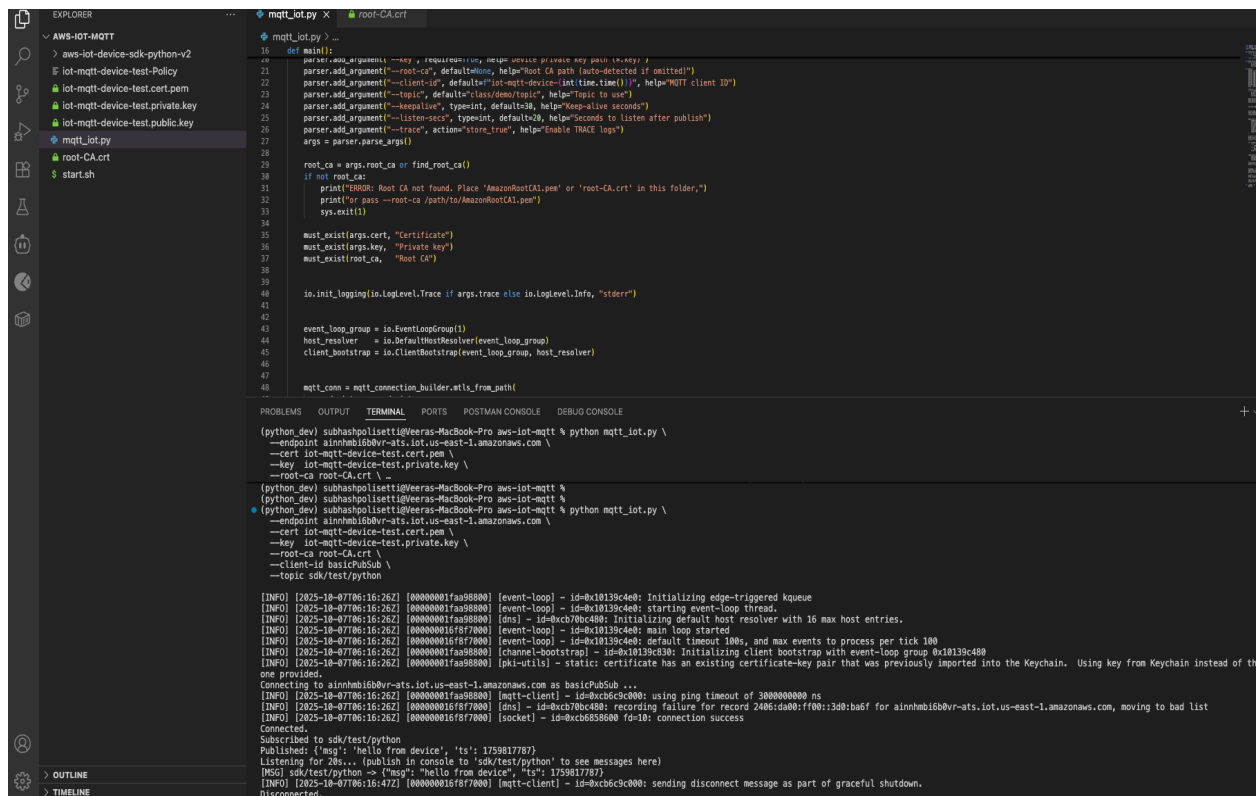
Installed aws IoT SDK in local machine

```
PROBLEMS  OUTPUT  TERMINAL  PORTS  POSTMAN CONSOLE  DEBUG CONSOLE
• (python_dev) subhashpoliseti@Veeras-MacBook-Pro aws-iot-mqtt % pip install awsiotsdk
Collecting awsiotsdk
  Downloading awsiotsdk-1.26.0-py3-none-any.whl.metadata (5.1 kB)
Collecting awscrt==0.28.1 (from awsiotsdk)
  Downloading awscrt-0.28.1-cp313-ab13-macosx_10_15_universal2.whl.metadata (7.3 kB)
Collecting awsiotsdk-1.26.0-py3-none-any.whl (80 kB)
  Downloading awsiotsdk-1.26.0-py3-none-any.whl (80 kB)
Collecting awscrt-0.28.1-cp313-ab13-macosx_10_15_universal2.whl (2.4 MB)
  Downloading awscrt-0.28.1-cp313-ab13-macosx_10_15_universal2.whl (2.4 MB)
Installing collected packages: awscrt, awsiotsdk
Successfully installed awscrt-0.28.1 awsiotsdk-1.26.0
• (python_dev) subhashpoliseti@Veeras-MacBook-Pro aws-iot-mqtt %
```



```
1 from mqttnot import mqttnot
2 from mqttnot import mqttnot_connection_builder
3 import json, time
4
5 # ENDPOINT = "YOUR_ENDPOINT_HERE"
6 # CLIENT_ID = "demo-thing-yourname"
7 # PATH_CERT = "demo-certificate.pem.crt"
8 # PATH_KEY = "private.pem.key"
9 # PATH_ROOT = "AmazonRootCA1.pem"
10 # TOPIC = "class/demo/topic"
11
12
13 ENDPOINT = "aiothmbi6b0vr-ats.iot.us-east-1.amazonaws.com"
14 CLIENT_ID = "iot-mqtt-device-test"
15 PATH_CERT = "iot-mqtt-device-test.cert.pem"
16 PATH_KEY = "iot-mqtt-device-test.private.key"
17 PATH_ROOT = "AmazonRootCA1.pem"
18 TOPIC = "class/demo/topic"
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[INFO] [2025-10-07T06:16:47Z] [000000016f8f7000] [mqtt-client] - id=0xcb6c9c000:
sending disconnect message as part of graceful shutdown.
Disconnected.



The screenshot shows a VS Code editor with a file explorer on the left and a terminal window at the bottom. The file explorer shows a project named 'AWS-IOT-MQTT' with files like 'aws-iot-device-sdk-python-v2', 'iot-mqtt-device-test-policy', 'iot-mqtt-device-test-cert.pem', 'iot-mqtt-device-test-private.key', 'iot-mqtt-device-test-public.key', 'mqtt_client.py', 'root-CA.crt', and 'start.sh'. The 'mqtt_client.py' file is open in the editor, showing a Python script that sets up an MQTT client. The terminal window shows the command 'python mqtt_client.py' being executed, and the output shows the client connecting to the AWS IoT Core endpoint, publishing a message, and then sending a disconnect message as part of a graceful shutdown.

```
def main():
    parser.add_argument("--key", required=True, help="MQTT client private key path (x.key)")
    parser.add_argument("--root-ca", default=None, help="Root CA path (auto-detected if omitted)")
    parser.add_argument("--client-id", default="iot-mqtt-device-test", help="MQTT client ID")
    parser.add_argument("--topic", default="class/demo/topic", help="Topic to use")
    parser.add_argument("--keepalive", type=int, default=30, help="Keep-alive seconds")
    parser.add_argument("--listen-sec", type=int, default=30, help="Seconds to listen after publish")
    parser.add_argument("--trace", action="store_true", help="Enable TRACE logs")
    args = parser.parse_args()

    root_ca = args.root_ca or find_root_ca()
    if not root_ca:
        print("ERROR: Root CA not found. Place 'AmazonRootCA1.pem' or 'root-CA.crt' in this folder.")
        print("For pass --root-ca /path/to/AmazonRootCA1.pem")
        sys.exit(1)

    must_exist(args.cert, "Certificate")
    must_exist(args.key, "Private key")
    must_exist(root_ca, "Root CA")

    io.init_logging(io.LogLevel.Trace if args.trace else io.LogLevel.Info, "stderr")

    event_loop_group = io.EventLoopGroup(1)
    host_resolver = io.DefaultHostResolver(event_loop_group)
    client_bootstrap = io.ClientBootstrap(event_loop_group, host_resolver)

    mqtt_conn = mqtt_connection_builder.mtls_from_path(

(python_dev) subhashpoliseti@Veras-MacBook-Pro aws-iot-mqtt % python mqtt_client.py \
--endpoint ainhmb16bvr-ats.iot.us-east-1.amazonaws.com \
--cert iot-mqtt-device-test-cert.pem \
--key iot-mqtt-device-test-private.key \
--root-ca root-CA.crt \
(python_dev) subhashpoliseti@Veras-MacBook-Pro aws-iot-mqtt % python mqtt_client.py \
--endpoint ainhmb16bvr-ats.iot.us-east-1.amazonaws.com \
--cert iot-mqtt-device-test-cert.pem \
--key iot-mqtt-device-test-private.key \
--root-ca root-CA.crt \
--client-id basicPubSub \
--topic sdk/test/python
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [event-loop] - id=0x18139c4e0: Initializing edge-triggered kqueue
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [event-loop] - id=0x18139c4e0: starting event-loop thread.
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [dns] - id=0xcb70bc480: Initializing default host resolver with 16 max host entries.
[INFO] [2025-10-07T06:16:26Z] [000000016f8f7000] [event-loop] - id=0x18139c4e0: main loop started
[INFO] [2025-10-07T06:16:26Z] [000000016f8f7000] [event-loop] - id=0x18139c4e0: default timeout 100s, and max events to process per tick 100
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [channel-bootstrap] - id=0x18139c480: Initializing client bootstrap with event-loop group 0x18139c4e0
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [skl-utils] - static: certificate has an existing certificate-key pair that was previously imported into the Keychain. Using key from Keychain instead of the one provided.
Connecting to ainhmb16bvr-ats.iot.us-east-1.amazonaws.com as basicPubSub...
[INFO] [2025-10-07T06:16:26Z] [00000001fa988800] [mqtt-client] - id=0xcb6c9c000: using ping timeout of 3000000000 ns
[INFO] [2025-10-07T06:16:26Z] [000000016f8f7000] [dns] - id=0xcb70bc480: recording failure for record 2400:da00:ff00:3d0:ba6f for ainhmb16bvr-ats.iot.us-east-1.amazonaws.com, moving to bad list
[INFO] [2025-10-07T06:16:26Z] [000000016f8f7000] [socket] - id=0xcb6858600 fd=10: connection success
Connected.
Subscribed to sdk/test/python
Published: {'msg': 'hello from device', 'ts': 1759817787}
Listening for 20s... (publish in console to 'sdk/test/python' to see messages here)
[MSG] sdk/test/python -> {'msg': 'hello from device', 'ts': 1759817787}
[INFO] [2025-10-07T06:16:47Z] [000000016f8f7000] [mqtt-client] - id=0xcb6c9c000: sending disconnect message as part of graceful shutdown.
Disconnected.
```

We created a Thing, policy, and certificates, and configured a Python-based client to connect securely to AWS IoT Core.